TELECOMMUNICATIONS · CONCEPTUAL REFERENCE

Mobile Network Protocol Stack Architecture

DATA JOURNEY
Application data → Radio signals → Network

ORIGIN DEVICE Mobile Station (MS)	LOGICAL COMMUNICATION	RECEIVING SYSTEM Base Station / Network
Application Layer User-facing network services · HTTP · DNS	PEER DATA	**Application Layer** Processes corresponding protocol data · 01
Transport Layer End-to-end data delivery · TCP · UDP	PEER DATA	**Transport Layer** Processes corresponding protocol data · 02
Network Layer Addressing and packet routing · IP	PEER DATA	**Network Layer** Processes corresponding protocol data · 03
Data Link Layer Framing and link access · Ethernet · Cellular link protocols	PEER DATA	**Data Link Layer** Processes corresponding protocol data · 04
Physical Layer Radio signal transmission · RF · Modulation	PEER DATA	**Physical Layer** Processes corresponding protocol data · 05

CONCEPT NOTE GSM, LTE, and 5G use technology-specific protocol layers and interfaces. This diagram is a simplified conceptual model.

LAYER ANNOTATIONS

What each layer does

1. **Application Layer**
   Provides services for web browsing, messaging, and mobile applications.
2. **Transport Layer**
   Supports reliable communication, flow control, and application data delivery.
3. **Network Layer**
   Handles logical addressing and routes packets between connected networks.
4. **Data Link Layer**
   Manages framing, medium access, and error detection over a single link.
5. **Physical Layer**
   Transmits raw bits as wireless radio signals using radio hardware.

LEGEND

- Bidirectional peer-layer communication
- Color identifies each protocol layer
- CLICK — Open a layer detail card

Simplified cellular communication path

Physical transmission proceeds through the radio access network to the core

Mobile Station (User equipment) → WIRELESS RADIO LINK → Base Station (Cell site) → ACCESS LINK → Radio Access (RAN) → BACKHAUL → Core Network (Services & routing)